TA-14 AUTHORITY GOVERNANCE INSTITUTION

Schillings Air × TA-14

Proposed real-time admissible execution governance for measurement-led HVAC retrofits

Technical exploration and bounded paid-pilot proposal | 29 October 2026

Prepared by Greggory Don Butler, Founder, TA-14 Authority Governance Institution

GOVERN THE DECISION. PROVE THE OUTCOME.

Schillings Air can continue to own its engineering, retrofit delivery, instrumentation, controls and customer relationship. TA-14 proposes a separate governance layer that determines whether a particular consequential action may be relied upon at the moment it is proposed, records the decision boundary, and examines the result afterward.

This document describes a proposed integration architecture, not an existing Schillings Air deployment. No Schillings Air system interfaces, contractual terms, sensor measurements, commercial commitments, or integration approvals have been independently verified for this proposal.

Governing question

Does the proposed consequence have sufficient admissible evidence, applicable authority, and established standing to become reality now?

Institutional rule

No admissible evidence. No admissible execution.

CONFIDENTIAL DISCUSSION DRAFT — do not distribute as an endorsed partnership announcement.

1. Executive decision brief

A building management system can measure, trend, alarm and actuate. A retrofit team can engineer a response. Those functions are essential, but they do not, by themselves, establish whether a specific proposed action is supported by admissible evidence, permitted by current authority and still valid when executed.

TA-14 is intended to examine that missing decision boundary without replacing the existing operational stack. The practical commercial hypothesis is that governance can reduce undocumented intervention, preserve decision lineage, expose evidence gaps and make post-retrofit claims more defensible.

What is proposed

- Observe: ingest bounded, attributable measurements and service records from agreed interfaces.
- Examine: test provenance, quality, freshness, continuity, thresholds, context, authority and permitted scope.
- Determine: return ALLOW, HOLD, DENY or ESCALATE with reason codes and evidence references.
- Bind: revalidate immediately before a consequential command or authorized work step.
- Preserve: capture action identity, timing, exceptions and outcome evidence in a replayable record.

What is not proposed

- No replacement of Schillings Air engineering judgment, building controls or established safety interlocks.
- No assertion that all sensor readings are automatically admissible, or that a green air-quality score proves safety.
- No production actuation or machine-speed performance guarantee before interface, safety and latency validation.

2. Proposed system boundary

Layer	Existing / proposed responsibility	Governance significance
Physical building	AHU, ventilation, filtration, sensors, controllers and occupied spaces	Observed reality and action consequences
Schillings Air / site stack	Retrofit design, monitoring, BMS, maintenance and operational workflows	Source of telemetry, authorized proposals and field execution
Evidence adapter (proposed)	Read-only intake, timestamp normalization, source identity, units, quality flags, lineage	Creates bounded candidate evidence; does not imply admissibility
TA-14 runtime (proposed)	Continuity, admissibility, authority, standing, scope, binding and determination	Establishes a documented right to rely or fail-closed outcome
Execution interface (optional)	Approved command pathway or technician work authorization	Only after separate safety and integration acceptance
Outcome record	Post-action readings, equipment response, deviations, sign-off and exception state	Examines what actually occurred, not merely what was intended

Minimum input contract for one decision

A proposed event should identify site, asset, action, requester, role, timestamp, intended consequence, source references, units, quality flags, policy version, authority scope and requested validity window. Missing fields should be explicitly classified rather than silently inferred.

Minimum response contract

The runtime should return determination, reason codes, evidence references, authority reference, decision timestamp, expiry/revalidation conditions, and immutable decision-record identifier. This is an illustrative contract, not a claim about an implemented Schillings Air API.

3. The eight-link admissible execution chain

Link	Required question	Potential evidence
1. Reality	What condition actually existed?	Sensor observation, asset state, occupancy and operating context
2. Record	Was it captured faithfully and attributable?	Source ID, calibration status, timestamp, unit, record identity
3. Continuity	Was custody and temporal continuity preserved?	Sequence, gap analysis, changes, corrections and provenance
4. Admissibility	Is this evidence fit for this decision?	Quality, relevance, freshness, threshold and limitations
5. Binding	Does current authority cover this action now?	Role, permit, policy, scope, expiry and current site state
6. Commit	Was a bounded decision authorized?	Determination, reason, approved parameters and record digest
7. Execution	What action actually occurred and by whom?	BMS command / work order, actor, time, actual settings
8. Outcome	What did the building do afterward?	Post-action trends, exceptions, verification and unresolved uncertainty

A completed record is valid only within its stated evidence window and scope. A changed sensor state, authority, policy or proposed action can require a new determination. The outcome record does not retroactively authorize an intervention that lacked valid standing at execution time.

4. Determination logic and operational safeguards

Disposition	Meaning	Illustrative retrofit condition
ALLOW	Evidence and current authority support the exact bounded action.	A validated ventilation adjustment within a permitted envelope.
HOLD	Evidence is incomplete, stale or not admissible yet.	CO_2 sensor quality flag fails or timestamp is outside the accepted window.
DENY	A governing prohibition or lack of authority bars the action.	Requested override exceeds an approved equipment or role boundary.
ESCALATE	A designated accountable authority must resolve a conflict.	Conflicting operational and infection-control requirements.

Fail-closed is not the same as unsafe shutdown

A governance HOLD or DENY must not be interpreted as permission to interrupt a safety-critical function. Emergency ventilation, fire/smoke control, medical or other protected sequences require their own engineering and regulatory treatment. The pilot must define which commands are in scope, which are observational only, and which must bypass the experimental gate under documented authority.

Runtime checks

- Verify the evidence window and policy version at the decision point, and again at binding.
- Reject unrecognized actors, unsupported command types and unauthorized scope expansion.
- Record any bypass, manual override, lost connectivity, changed context or expired authorization.
- Treat latency and availability as measurable acceptance criteria, not marketing assertions.

5. Worked example — ventilation intervention

Scenario: An occupied zone shows elevated CO_2 during a scheduled operating period. A proposed control action would increase outside-air ventilation. All values and actions below are hypothetical and are not Schillings Air measurements.

Stage	Proposed record	Decision issue
Pre-action baseline	Zone CO_2 trend, outdoor-air condition, airflow, occupancy, filter status and AHU operating mode	Are readings relevant, recent, attributable and internally consistent?
Authority	Requester identity, BMS point permissions, operating policy, allowable setpoint range	Is the actor authorized for this exact adjustment now?
Determination	ALLOW / HOLD / DENY / ESCALATE with bounded settings and expiry	Can the action be relied on without inference?
Intervention	Commanded change, actual BMS acknowledgment, technician or controller identity	Did execution match the authorized commit?
Post-action	CO_2 response, airflow, energy or comfort side effects, alarms and exception flags	Was the intended outcome supported, limited or not established?

Illustrative HOLD branch

If the CO_2 signal is stale or has an unresolved quality flag, the proposed ventilation change returns HOLD rather than converting a plausible reading into an authorized consequence. The site can collect new evidence or escalate under its approved operational procedure.

Illustrative ALLOW branch

If evidence, scope and authority pass the bounded checks, the runtime may issue an expiring ALLOW for a specified adjustment. The record must still distinguish command issued, command acknowledged, actual physical response and verified environmental outcome.

6. HVAC evidence-before-intervention workflow

TA-14’s HVAC-specific sequence is compatible with a disciplined retrofit evaluation, but must be mapped to Schillings Air’s own technical procedures rather than imposed as a substitute.

Phase	Purpose	Deliverable
HPR-BL baseline	Document equipment and environmental performance before intervention.	Bounded pre-action record and source-quality statement.
NIRET gate	Determine whether an invasive or specialist step is warranted and authorized.	Intervention eligibility and scope determination.
DDD	Issue a Declared Diagnostic Determination only after the required baseline is complete and frozen.	Traceable diagnostic conclusion, limitations and standing.
Authorized intervention	Execute only the justified work under applicable site authority.	Bounded action, identity, timing and exceptions.
HPR-V verification	Re-examine performance after the work when applicable.	Outcome record and unresolved issues.

Routine maintenance may require no invasive intervention. The governance objective is to prevent a measurement, recommendation or commercial incentive from being mistaken for authorization to act.

7. Evidence and security design requirements

Data integrity

- Document system-of-record ownership, sensor and BMS identifiers, units, calibration metadata, time synchronization, retention, corrections and missing data.
- Separate raw observations from derived scores, alerts, recommendations and authoritative determinations.
- Preserve evidence references and hashes where technically and legally appropriate; do not imply that hashing alone proves measurement truth.

Authority and control

- Define role-based access, accountable owner, change approval, command allowlist and site-specific authority source.
- Support revocation, expiry and revalidation at the bind point; document emergency and manual override rules.
- Isolate pilot credentials, use least privilege and avoid write access during shadow mode.

Evaluation and acceptance

- Test stale sensors, contradictory inputs, clock drift, policy changes, lost network, revoked authority, duplicate commands and attempted bypass.
- Measure decision latency, error rate, false holds, replay completeness and outcome-record integrity against agreed thresholds.
- Agree in advance what evidence would justify advancing from read-only evaluation to any live gate.

8. Paid pilot — scope, deliverables and acceptance

Phase A: bounded, read-only shadow evaluation

Select one facility, one AHU or equivalent asset class, and one narrow decision such as a ventilation adjustment recommendation. Use approved historical and live read-only data. Map source quality, operational policy, authority owners, action proposals and exceptions. Run scenarios without actuating equipment.

Phase A deliverables

- Integration architecture and field-level data dictionary.
- Decision policy and authority matrix for the selected use case.
- Recorded ALLOW, HOLD, DENY and ESCALATE scenario outcomes, including failure cases.
- Replayable sample evidence-to-decision records and limitations register.
- Written recommendation on whether a controlled next phase is technically justified.

Phase B: optional controlled execution

A separate authorization and statement of work would be required before any production-connected gate. Preconditions include engineering review, security review, safety-interlock boundaries, measured latency and availability, rollback procedure, named accountable authorities and signed acceptance criteria.

Commercial structure

This is a paid pilot proposal, not a free integration commitment. Pricing, duration, access, ownership, confidentiality, liability, support and success metrics should be agreed in a mutually executed statement of work. No fee, schedule or guaranteed savings is asserted here.

9. Decision meeting and requested materials

The immediate objective is a technical scoping conversation, not a public partnership announcement.

From Schillings Air / site team	From TA-14
Solutions overview, target building type and retrofit workflow	Bounded governance method and technical decision schema
Available BMS/sensor interfaces and sample anonymized event payloads	Evidence adapter requirements and integration options
Named site authority, policy constraints and safety boundaries	Authority, standing, determination and revalidation model
Candidate pilot asset, decision and measurable objectives	Proposed pilot test plan, failure cases and outcome record

Proposed working-session agenda

- 15 minutes: Schillings Air system boundaries and existing decision process.
- 15 minutes: TA-14 event-to-determination example and evidence requirements.
- 15 minutes: select one bounded shadow-mode use case and identify authority owners.
- 15 minutes: pilot acceptance criteria, commercial scope and next document exchange.

Release and representation controls

Any public-facing showroom, case study, logo treatment, quoted result or announcement referencing Schillings Air should be approved by the appropriate company representative. This draft is prepared for discussion and does not represent an endorsement or executed partnership.

Greggory Don Butler
Founder, TA-14 Authority Governance Institution
https://www.ta14authority.org